Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A mass of $50\ \mathrm{g}$ of water in a closed vessel, with surroundings at a constant temperature takes 2 minutes to cool from $30^{\circ}\mathrm{C}$ to $25^{\circ}\mathrm{C}$. A mass of $100\ \mathrm{g}$ of another liquid in an identical vessel with identical surroundings takes the same time to cool from $30^{\circ}\mathrm{C}$ to $25^{\circ}\mathrm{C}$. The specific heat of the liquid is : (The water equivalent of the vessel is $30\ \mathrm{g}$.)

(A) $2.0\mathrm{kcal/kg}$

(B) $7\mathrm{kcal/kg}$

(C) $3\mathrm{kcal/kg}$

(D) $0.5\mathrm{kcal/kg}$

Q.2

Mass of water which absorbs or emits the same amount of heat as is done by the body for the same rise or fall in temperature is known as

(A) thermal capacity of the body

(B) specific heat capacity of the body

(C) latent heat capacity of the body

(D) water equivalent of the body

Q.3

A hammer of mass $1\ \mathrm{kg}$ having speed of $50\ \mathrm{m/s}$ hit a iron nail of mass $200\mathrm{gm}$. If specific heat of iron is $0.105\mathrm{cal/gm^{\circ}C}$ and half the energy is converted into heat. the raise in temperature of nail is

(A) $7.1^{\circ}\mathrm{C}$

(B) $9.2^{\circ}\mathrm{C}$

(C) $10.5^{\circ}\mathrm{C}$

(D) $12.1^{\circ}\mathrm{C}$

Q.4

In an energy recycling process, $100\ \mathrm{g}$ of steam $\mathrm{at}\ 100^{\circ}\mathrm{C}$ becomes water at $100^{\circ}\mathrm{C}$ which converts $\mathrm{y}\ g$ of ice at $0^{\circ}\mathrm{C}$ into water at $100^{\circ}\mathrm{C}$. The numeric value of y is

(A) 100

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(B) 200

(C) 300

(D) 400

Q.5

In a water-fall the water falls from a height of 100 m. If the entire K.E. of water is converted into heat, the rise in temperature of water will be

(A) $0.23^{\circ}\mathrm{C}$

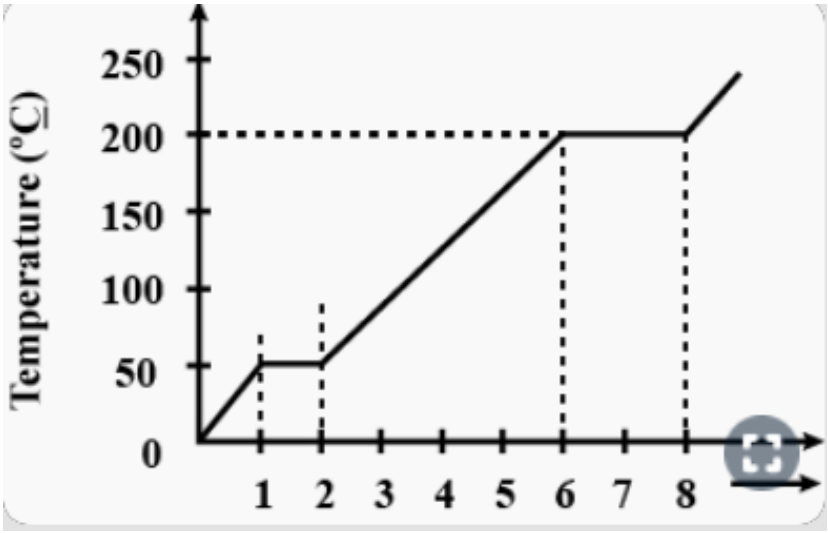
(B) $0.46^{\circ}\mathrm{C}$

(C) $2.3^{\circ}\mathrm{C}$

(D) $0.023^{\circ}\mathrm{C}$

Q.6

A student takes 50 gm wax (specific heat $= 0.6\mathrm{kcal/kg^{\circ}C}$) and heats it till it boils.
The graph between temperature and time is as follows.Heat supplied to the wax per minute and boiling point are respectively



(A) $500\mathrm{cal}, 50^{\circ}\mathrm{C}$

(B) $1000\mathrm{cal}, 100^{\circ}\mathrm{C}$

(C) $1500\mathrm{cal}, 200^{\circ}\mathrm{C}$

(D) $1000\mathrm{cal}, 200^{\circ}\mathrm{C}$

Q.7

The heat (Q) supplied to a solid, which is otherwise thermally isolated from its surroundings, is plotted as a function of its absolute temperature, θ. It is found that they are related by the equation. $Q = a\theta^2 + b\theta^4$. $(a, b$ are constants $)$
The heat capacity of the solid is given by

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(A) $a\frac{\theta^3}{3}+b\frac{\theta^5}{5}$

(B) $a\theta+b\theta^3$

(C) $a\frac{\theta}{3}+b\frac{\theta^3}{5}$

(D) $2a\theta+4b\theta^3$

Q.8

A lead bullet strikes against a steel plate with a velocity $200\ \mathrm{m\ s^{-1}}$. If the impact is perfectly inelastic and the heat produced is equally shared between the bullet and the target, then the rise in temperature of the bullet is (specific heat capacity of lead $=125\ \mathrm{J\ kg^{-1}\ K^{-1}}$)

(A) $80^\circ\mathrm{C}$

(B) $60^\circ\mathrm{C}$

(C) $160^\circ\mathrm{C}$

(D) $40^\circ\mathrm{C}$

Q.9

Steam at $100^\circ\mathrm{C}$ is passed into $20\ \mathrm{g}$ of water at $10^\circ\mathrm{C}$. When water acquires a temperature of $80^\circ\mathrm{C}$. Then what will be the mass (in gram) of water present? [Take specific heat of water $=1\mathrm{calg^{-1}{}^\circ C^{-1}}$ and latent heat of steam $=540\ \mathrm{cal\ g^{-1}}$]

(A) 22.7

(B) 22.3

(C) 22.9

(D) 22.5

Q.10

$4200\ \mathrm{J}$ of work is required for

(A) increasing the temperature of $10\ \mathrm{g}$ of water through $10^\circ\mathrm{C}$

(B) increasing the temperature of $100\ \mathrm{g}$ of water through $10^\circ\mathrm{C}$

(C) increasing the temperature of $1\ \mathrm{kg}$ of water through $10^\circ\mathrm{C}$

(D) increasing the temperature of $500\ \mathrm{g}$ of water through $10^\circ\mathrm{C}$

Q.11

The latent heat of vaporization of a substance is always

(A) greater than its latent heat of fusion

(B) greater than its latent heat of sublimation

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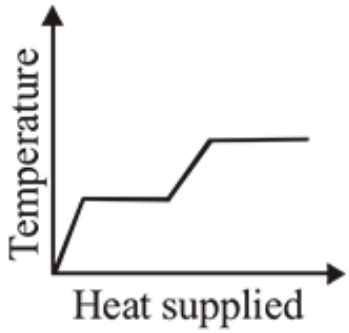
(C) equal to its latent heat of sublimation

(D) less than its latent heat of fusion

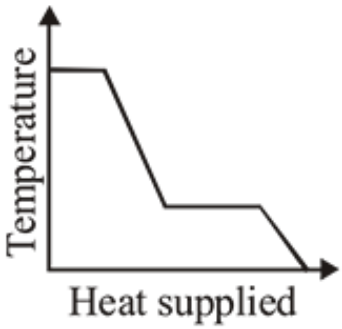
Q.12

A block of ice at $-10^{\circ}\mathrm{C}$ is slowly heated and converted to steam at $100^{\circ}\mathrm{C}$. Which of the following curves represents the phenomenon qualitatively

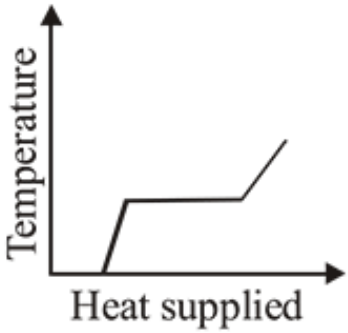
(A)



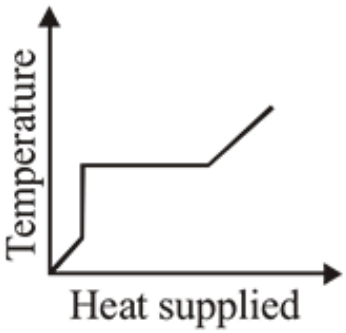
(B)



(C)



(D)



Q.13

$500\ \mathrm{g}$ of water and $100\ \mathrm{g}$ of ice at $0^{\circ}\mathrm{C}$ are in a calorimeter whose water equivalent is 40 g. $10\ \mathrm{g}$ of steam at $100^{\circ}\mathrm{C}$ is added to it. Then water in the calorimeter is : (Latent heat of ice $= 80\mathrm{cal/g}$, Latent heat of steam $= 540\mathrm{cal/g}$)

(A) $580\ \mathrm{g}$

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(B) $590\ \mathrm{g}$

(C) $600\ \mathrm{g}$

(D) $610\ \mathrm{g}$

Q.14

A beaker contains $200\mathrm{gm}$ of water. The heat capacity of the beaker is equal to that of $20\mathrm{gm}$ of water. The initial temperature of water in the beaker is $20^{\circ}\mathrm{C}$. If $440\mathrm{gm}$ of hot water at $92^{\circ}\mathrm{C}$ is poured in it, the final temperature, neglecting radiation loss, will be nearest to

(A) $58^{\circ}\mathrm{C}$

(B) $68^{\circ}\mathrm{C}$

(C) $73^{\circ}\mathrm{C}$

(D) $78^{\circ}\mathrm{C}$

Q.15

$2\ \mathrm{kg}$ of water is heated from $0^{\circ}\mathrm{C}$ to $4^{\circ}\mathrm{C}$. The specific heat capacity of water is $4200\ \mathrm{J/kgK}$ and its densities are $999.9\ \mathrm{kg/m^3}$ and $1000\ \mathrm{kg\ m^3}$ respectively at these temperatures. The atmospheric pressure is 10^5 pascal. What is the change in internal energy?

(A) $33.6\ \mathrm{kJ}$

(B) $\left(33.6-2\times10^{-5}\right)\ \mathrm{kJ}$

(C) $\left(33.6+2\times10^{-5}\right)\ \mathrm{kJ}$

(D) Zero

Q.16

Intemal energy of two moles of an ideal gas at a temperature of $127^{\circ}\mathrm{C}$ is $1200\mathrm{R}$. Then, the specific heat of the gas at constant pressure is

(A) $0.5\mathrm{R}$

(B) $0.1\mathrm{R}$

(C) $1.5\mathrm{R}$

(D) $2.5\mathrm{R}$

Q.17

The temperatures of equal masses of three different liquids A, B and C are $12^{\circ}\mathrm{C}$, $19^{\circ}\mathrm{C}$ and $28^{\circ}\mathrm{C}$, respectively. The temperature when A and B are mixed is $16^{\circ}\mathrm{C}$, while when B and C are mixed, it is $23^{\circ}\mathrm{C}$. What would be the temperature when A and C are mixed?

(A) 20.26 degree celsius

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(B) 30.26 degree celsius

(C) 21 degree celsius

(D) 40.26 degree celsius

Q.18

1 kg ice at $-20^{\circ}\mathrm{C}$ is mixed with 1 kg steam at $200^{\circ}\mathrm{C}$. Find the mass of water after mixing.

(A) 34/27 kg

(B) 35/24 kg

(C) 5 kg

(D) 4/23 kg

Q.19

1 g of ice at 0°C is into steam at 100°C. The heat supplied in this process is

(A) $100\ \mathrm{cal}$

(B) $0.01\ \mathrm{cal}^{\circ}\mathrm{C}^{-1}$

(C) $720\ \mathrm{cal}$

(D) $1\ \mathrm{kilocal}$

Q.20

A sample of 10 g of steam at $100^{\circ}\mathrm{C}$ with 6 g ice at $0^{\circ}\mathrm{C}$ is kept inside a calorimeter initially. Find the amount of steam in g left?

$L_{Fusion} = 80\ \mathrm{cal/g},\ L_{Vaporiration} = 540\ \mathrm{cal/g},$ Specific heat capacity $= 1\ \mathrm{cal/g}\ ^{\circ}\mathrm{C}$

(A) 7

(B) 9

(C) 6

(D) 8

Answers & Solutions

Q.1 Answer:

$0.5\mathrm{kcal/kg}$

Solution:

$\frac{(m_w C_w + W)\Delta T}{t} = \frac{(m_\ell C_\ell + W)\Delta T}{t}$

$(W = \textbf{water equivalent of the vessel})$ **or,** $m_w C_w = m_\ell C_\ell$

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$\therefore$ **Specific heat of liquid,** $C = \frac{m_W C_W}{m_\ell}$
$= \frac{50 \times 1}{100} = 0.5 \text{kcal/kg}$

Q.2 Answer:

water equivalent of the body

Solution:

Water equivalent is the quantity of water whose thermal capacity is same as the heat capacity of the body.

Q.3 Answer:

$7.1^\circ C$

Solution:

$W = JQ \Rightarrow \frac{1}{2}\left(\frac{1}{2}Mv^2\right) = J(m.c.\Delta\theta)$
$\frac{1}{4} \times 1 \times (50)^2 = 4.2[200 \times 0.105 \times \Delta\theta]$
$\Rightarrow \Delta\theta = 7.1^\circ C$

Q.4 Answer:

300

Solution:

Specific heat of water $= 4200\ J/kg - K$ **Latent heat of fusion** $= 3.36 \times 10^5\ J/kg$
Latent heat of vapourisation $= 22.68 \times 10^5\ J/kg$
$x \times 10^{-3} \times 22.68 \times 10^5\ J$
$= y \times 10^{-3} \times 3.36 \times 10^5\ J + y \times 10^{-3} \times 4200 \times 100$
$\therefore \frac{x}{y} = \frac{7.56}{22.68} = \frac{1}{3}$

Q.5 Answer:

$0.23^\circ C$

Solution:

$\Delta\theta = 0.0023\ h = 0.0023 \times 100 = 0.23^\circ C$

Q.6 Answer:

$1500 \text{cal}, 200^\circ C$

Solution:

Since specific heat $= 0.6 \text{kcal/gm} \times {}^\circ C = 0.6 \text{cal/gm} \times {}^\circ C$
From graph it is clear that in a minute, the temperature is raised from $0^\circ C$ **to** $50^\circ C$.
$\Rightarrow$ **Heat required for a minute** $= 50 \times 0.6 \times 50 = 1500$ **cal .**
Also from graph, Boiling point of wax is $200^\circ C$.

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Q.7 Answer:

$2a\theta + 4b\theta^3$

Solution:

$\varphi = a\theta^2 + b\theta^4\theta \rightarrow ab$ solute temperature const. $\theta \rightarrow$ **absolute temperature Const.** $c \rightarrow$ **heat ceepacity of the solid**

$$C = \int mCdT = \int cdT$$
$$d\varphi = cdT$$
$$c = \frac{d\varphi}{dT}$$
$$c = \frac{d}{d\theta}\left(a\theta^2 + b\theta^2\right)$$
$$c = 2a\theta + c^3$$
$$c = 2a\theta + 4b\theta^3$$

Q.8 Answer:

$80^\circ\mathrm{C}$

Solution:

$$\text{Heat produced} = \mathrm{ms}\Delta\mathrm{T} = \frac{1}{2}\left(\frac{1}{2}\mathrm{mv}^2\right)$$
$$\Rightarrow \Delta\mathrm{T} = \frac{\mathrm{v}^2}{4\,\mathrm{s}} = \frac{(200)^2}{4\times125} = \frac{4\times10^4}{4\times125} = 80^\circ\mathrm{C}$$

Q.9 Answer:

22.5

Solution:

According to the principle of calorimetry. Heat lost = Heat gained $\mathrm{mL_v} + \mathrm{ms_w}\Delta\theta = \mathrm{m_w s_w}\Delta\theta$

$\Rightarrow \mathrm{m} \times 540 + \mathrm{m} \times 1 \times (100 - 80)$

$= 20 \times 1 \times (80 - 10)$

$\Rightarrow \mathrm{m} = 2.5\ \mathrm{g}$

Therefore total mass of water at $80^\circ\mathrm{C}$

$= (20 + 2.5)\mathrm{g} = 22.5\ \mathrm{g}$

Q.10 Answer:

increasing the temperature of $100\ \mathrm{g}$ of water through $10^\circ\mathrm{C}$

Solution:

Work done to raise the temperature of 100 gm **water through** $10^\circ\mathrm{C}$ **is** $\mathrm{W} = \mathrm{JQ} = 4.2 \times \left(100 \times 10^{-3} \times 1000 \times 10\right)$

$= 4200\ \mathrm{J}$

Q.11 Answer:

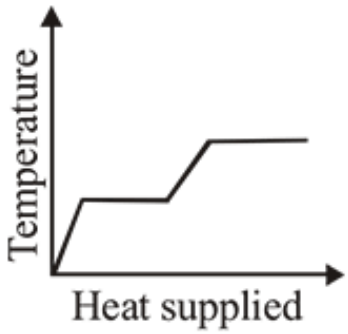
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greater than its latent heat of fusion

Solution:

The latent heat of vaporization is always greater than latent heat of fusion because in liquid to vapour phase change there is a large increase in volume. Hence more heat is required as compared to solid to liquid phase change.

Q.12 Answer:



Solution:

Initially, on heating temperature rises from $-100^{\circ}\mathrm{C}$ to $0^{\circ}\mathrm{C}$. then ice melts and temperature does not rise. After the whole ice has melted, temperature beigins to rise until it reaches $100^{\circ}\mathrm{C}$. then it becomes constant, as at the boiling point will not rise.

Q.13 Answer:

$590\ \mathrm{g}$

Solution:

As $1\ \mathrm{g}$ of steam at $100^{\circ}\mathrm{C}$ melts $8\ \mathrm{g}$ of ice at $0^{\circ}\mathrm{C}$.
$10\ \mathrm{g}$ of steam will melt $8 \times 10\ \mathrm{g}$ of ice at $0^{\circ}\mathrm{C}$ Water in calorimeter $= 500 + 80 + 10\ \mathrm{g} = 590\ \mathrm{g}$

Q.14 Answer:

$68^{\circ}\mathrm{C}$

Solution:

Let the final temperature be T. Then $200 \times 1 \times (\mathrm{T} - 20) + 20 \times (\mathrm{T} - 20)$
$= 440(92 - \mathrm{T})$
Solving it, we get $\mathrm{T} = 68^{\circ}\mathrm{C}$.

Q.15 Answer:

$\left(33.6 - 2 \times 10^{-5}\right)\ \mathrm{kJ}$

Solution:

$\mathrm{m} = 2\mathrm{Kg} \quad \mathbf{T}_1 = 0^{\circ}\mathrm{C} \quad \mathbf{T}_2 = 4^{\circ}\mathrm{C} \quad \mathrm{s} = 420\ \mathrm{J/KgK}$
density $\rho_1 = 999.9\ \mathrm{kg/m^3}$ and $\rho_2 = 1000\ \mathrm{kg/m^3}$

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$\mathrm{P} = 10^5$ **Pascal**

Internal energy of gass is

$\Delta \mathbf{U} = \Delta \mathrm{Q} - \Delta \mathrm{W}$

$\Delta Q = m\Delta t = 2 \times 4200(4-0) \quad = 33.6\ \mathrm{kJ}$

Work done $\Delta \mathrm{W} = \mathrm{p}\Delta \mathrm{V}$

Since the volume = mass $\times$ **density**

$= \mathrm{mP}\left(\frac{1}{\rho_2} - \frac{1}{\rho_1}\right) \quad = 2 \times 10^5 \left(\frac{1}{1000} - \frac{1}{999.9}\right) = 2 \times 10^{-5}\ \mathrm{kJ}$

Hence $\Delta \mathrm{U} = \Delta \mathrm{Q} - \Delta \mathrm{W}$

$= \left(33.6 - 2 \times 10^{-5}\right)\ \mathbf{kJ}$

Q.16 Answer:

$2.5\mathrm{R}$

Solution:

Internal energy of 2 mole of gas is

$\Delta \mathrm{U} = 1200\mathrm{R}$ **at** $\mathrm{T} = 127^{\circ}\mathrm{C} = 400\ \mathrm{K}$

$\Delta \mathrm{U} = \mathrm{nCv}\Delta \mathrm{T}$

Specific heat at constant volume

$\mathrm{Cv} = \frac{1200\mathrm{R}}{2 \times 400} = 1.5\mathrm{R}$

as we know

$\mathrm{Cp} - \mathrm{Cv} = \mathrm{R}$

$Cp = Cv + R$

$= 1.5\mathbf{R} + \mathbf{R}$

$= 2.5\mathrm{R}$

Q.17 Answer:

20.26 degree celsius

Solution:

Let m **= mass of each liquid, when** A **and** B **are mixed.**

Heat lost by B **= heat gained by** A

$\Rightarrow ms_B(19-16) = ms_A(16-12) \Rightarrow 3s_B = 4s_4$ **...(i)**

When B **and** C **are mixed,**

Heat lost by C **= heat gained by** B

$\Rightarrow ms_C(28-23) = ms_B(23-19) \Rightarrow 5s_C = 4s_B$ **...(ii)**

From Eqs. (i) and (ii), we get

$16s_A = 12s_B = 15s_C$ **...(iii)**

When A **and** C **are mixed, Let** θ **= tinal temperature**

Heat lost by C **= heat gained by** A

$\Rightarrow \quad ms_(28-\theta) = ms_A(\theta - 12)$

Using Eq. (iii), we get $\Rightarrow \quad 15s_6(28-\theta) = 15s_4(\theta - 12)$

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$\Rightarrow \quad 16x_A(28-\theta) = 15s_t(\theta - 12)$

On solving for θ, we get $\theta = \frac{16\times28+12\times15}{16+15}$

$\Rightarrow \quad \theta = 20.26^\circ\text{C}$

Q.18 Answer:

34/27 kg

Solution:

Let equilibrium temperature be 100°C, heat required to convert 1 kg ice at -20°C to 1 kg water at 100°C is equal to

$H_1 = 1 \times \frac{1}{2} \times 20 + 1 \times 80 + 1 \times 1 \times 100 = 190\text{kcal}$

Heat released by steam to convert 1 kg steam at 200°C to 1 kg water at 100°C is equal to

$H_2 = 1 \times \frac{1}{2} \times 100 + 1 \times 540 = 590\text{kcal}$

1kg ice at $-20^\circ\text{C} = H_1 + 1$ kg water at 100°C ...(i)

1 kg steam at $200^\circ\text{C} = H_2 + 1$ kg water at 100°C ...(ii)

By adding Eqs. (i) and (ii)

1 kg ice at $-20^\circ\text{C} + 1$ kg steam at $200^\circ\text{C} = H_1 + H_2 + 2$ kg water at 100°C.

Here heat required to ice is less than heat supplied by steam, so mixture equilibrium temperature is 100°C. Then steam is not completely converted into water.

So mixture has water and steam, which is possible only at 100°C. Mass of steam which converted into water is equal to

$m = \frac{190-1\times\frac{1}{2}\times100}{540} = \frac{7}{27}$ kg

So mixture content Mass of steam $= 1 - \frac{7}{27} = \frac{20}{27}$ kg

Mass of water $= 1 + \frac{7}{27} = \frac{34}{27}$ kg

Q.19 Answer:

720 cal

Solution:

Using the formula,

1) $Q = m \times s \times \Delta T$,

where Q, s **and** ΔT **are heat required, specific heat capacity and change in temperature respectively.**

2) $Q = mL$,

where m **and** L **are mass and latent heat respectively.**

Values of latent heat of fusion and vapourization are 80 cal / g **and** 540 cal / g**, while specific heat of water is** 1 cal / kg /$^\circ$ C**.**

Total heat required must be $= \text{m} \times 80 + \text{m} \times 1 \times 100 + \text{m} \times 540$

$= \text{m}(720) = 1 \times 720 = 720$ cal

Q.20 Answer:

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Solution:

$$\underbrace{6\times 80}_{\text{fusion}}+\underbrace{6\times 1\times 100-0}_{\text{raise temp}}=\underbrace{x\times 540}_{\text{condensation}}$$

$$\therefore \quad x=2\text{ g} \quad \Rightarrow \quad \text{left steam}=8 \quad \text{g}$$

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